

Homework 6 Solutions

1. Prove Case 2 of Theorem 13.2 above.

Solution: In this case, we have $\deg(x) \geq 3$, meaning there are at least 3 vertices adjacent to x , say a, b, c . So xa, xb , and xc are all edges. Now note that either there's an edge between two of the elements $\{a, b, c\}$ (in which case they form a clique with x) or there are no edges between any of them, in which case they form an independent set.

2. Suppose G is a connected graph. Show that between every pair of distinct vertices x, y of G , there is a simple path connecting x to y in G . (hint: consider a shortest path between x and y and use proof by contradiction.)

Solution: Suppose there is a path from x to y . The well-ordering principle tells us that there is a *shortest* path. Suppose the shortest path between x and y is not simple, i.e. it repeats the element a . Then it looks like $x, b_1, \dots, b_k, a, c_1, \dots, c_m, a, d_1, \dots, d_n, y$. But then the path $x, b_1, \dots, b_k, a, d_1, \dots, d_n, y$ is a shorter path. Contradiction!

3. The *degree sequence* of a graph is a list of the degrees of its vertices in nonincreasing order. Prove that if a simple graph has degree sequence 2, 2, 2, 1, 1, 1, 1 then it must be disconnected.

Solution: Let G be a simple connected graph with the above degree sequence. Let a be a vertex in G with degree 2. We know a is adjacent to two other vertices, say b and c . If both b and c have degree 1, then they cannot be adjacent to anything else. So any path from a to any other vertex can only contain b or c . But there are other vertices in G besides a, b, c ; thus G is disconnected, which is a contradiction. So we must have that either b or c has degree 2. WLOG let b have degree 2. Then either b is adjacent to c (in which case the graph is disconnected like before) or b is adjacent to some d not among a, b, c . Now if c and d both have degree 2, we get into a contradiction again just like before; hence, either c or d has degree 1. WLOG we can assume c has degree 1 (if not, just repeat this argument switching the roles of a and b). So d has degree 2, meaning it is only adjacent to b . And there must be an e adjacent to c , which has degree 1 and so is only adjacent to c . But then, any path from a to any other vertex can only contain a, b, c, d , or e . But there are other vertices in G with degree 1; hence G is disconnected. Contradiction!

4. For $n \geq 2$, define the cycle graph $C_n = (V, E)$ by $V = \{0, \dots, n-1\}$ and $\{x, y\} \in E \Leftrightarrow x - y \equiv \pm 1 \pmod n$. Prove that for every $n \geq 2$, the graph C_n is connected.

Solution: Consider any two distinct $x, y \in V$ and WLOG assume $x < y$. Then there is a path $x, x+1, \dots, y$ connecting x to y . Hence, C_n is connected.

5. Prove Theorem 14.1 above: every graph is a disjoint union of connected components.

Solution: We use induction on the cardinality of the vertex set V .

Base case: $|V| = 1$. There is just one graph with only one vertex, and it is connected, so it is a disjoint union of connected components.

Inductive step: Suppose that every graph $G = (V, E)$ of size k is a disjoint union of connected components. Let G_0 be a graph whose vertex set V_0 has cardinality $k + 1$. Pick some $a \in V_0$. Then $V_0 = \{a\} \cup (V_0 \setminus \{a\})$. Either there is no edge between a and any vertex in $V_0 \setminus \{a\}$, or there is an edge between a and something else. If there is no edge, then $\{a\}$ is a connected component, and any other connected component of G_0 will not contain a . Hence, we can write V_0 as $\{a\} \cup \bigcup_i (V_0 \setminus \{a\})_i$ where the $(V_0 \setminus \{a\})_i$ s are the connected components of $V_0 \setminus \{a\}$.

Now consider the case where there is an edge between a and some other vertex in $V_0 \setminus \{a\}$. Let $\{(V_0 \setminus \{a\})_{i_1}, \dots, (V_0 \setminus \{a\})_{i_q}\}$ enumerate the connected components of $V_0 \setminus \{a\}$ which contain a vertex adjacent to a . Then for any $x, y \in S = \{a\} \cup \bigcup_{j < q} (V_0 \setminus \{a\})_{i_j}$, there is a path from x to y (potentially using a if they are in different connected components). Hence, S is connected. Furthermore, S is still disconnected from any other connected components of $V_0 \setminus \{a\}$ since any path either uses a (but there are no edges from a to those components) or does not use a (in which case there are no paths between elements of S and other components). So we can write V_0 as $S \cup \bigcup_{j < r} (V_0 \setminus \{a\})_{i_j}$, where the $(V_0 \setminus \{a\})_{i_j}$ s enumerate the connected components of V_0 not contained in S .

6. Show that for every graph G , if G is not connected then its complement must be connected.

Solution: Suppose $G = (V, E)$ is not connected. Then it can be split into at least two distinct connected components. Let x, y be any two vertices in V . If x, y are in different connected components, then there is an edge in $\overline{G}$ between x and y . If x, y are in the same connected component, we can pick some a in a different connected component. Then we have edges xa and ay in $\overline{G}$, and so there is a path between x and y .

7. Prove that C_n is 2-colorable iff n is even.

Solution: Suppose n is even. Then we can 2-color C_n by making all the evens blue and all the odds red. Since for each $x \in \{0, \dots, n-1\}$, if $0 < x < n-1$ then x is only adjacent to numbers of different parity than x , and 0 and $n-1$ are colored differently, this is a valid 2-coloring.

Now suppose n is odd. Suppose there is a 2-coloring of C_n . WLOG assume 0 is colored red. Then inductively we can see that all the evens are colored red and all the odds are colored blue. Since $n-1$ is even, $n-1$ is colored red. But $n-1$ is adjacent to 0; contradiction!

8. Show that the hypercube graph H_n is connected and 2-colorable.

Solution: To show that H_n is connected, we must show there exists a path between any two vertices. So let $\vec{a} = (a_1, \dots, a_n)$ and $\vec{b} = (b_1, \dots, b_n)$ be any two vertices on the hypercube. Considering each i from 1 to n in sequence, change each entry of a_i one-by-one to match b_i if it disagrees. Each of these changes describes an edge in the hypercube. Listing these edges and modified versions of a in a sequence gives a path from a to b .

To see that H_n is 2-colorable, we color each vertex $\vec{a} = (a_1, \dots, a_n)$ red if it has an even number of 1s and blue if it has an odd number of 1s. To see that this works, note that if $(a_1, \dots, a_n)$ is adjacent to $(b_1, \dots, b_n)$ and $(a_1, \dots, a_n)$ contains an even number of 1s, then since exactly one of the $b_1, \dots, b_n$ s is different; hence, it either flips a 1 to a 0 or a 0 to a 1. In either case it has an odd number of 1s.